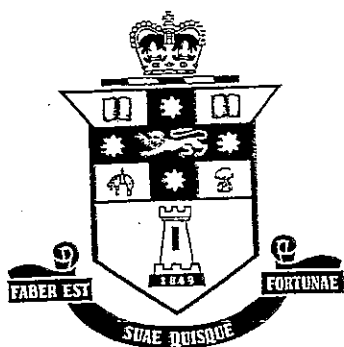




Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2006

**HIGHER SCHOOL CERTIFICATE COURSE
ASSESSMENT TASK 3**

Mathematics Extension 1

TIME ALLOWED: 1 HOUR

Outcomes Assessed	Questions	Marks
Manipulates algebraic expressions and calculus involving logarithmic and exponential functions..	1	
Determines integrals by reduction to a standard form through a given substitution	2	
Uses the relationship between functions, inverse functions and their derivatives.	3	
Evaluates mathematical solutions to problems and communicates them in appropriate form.	All	

Question	1	2	3	Total	%
Marks	/17	/17	/19	/53	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started on a new page

Question 1 (17 marks)

a) Show that $\frac{e^x}{e^x + e^{-x}} = \frac{e^{2x}}{e^{2x} + 1}$ [2]

b) The curves $y = \frac{2e}{x}$ and $y = \log_e x^2$ intersect at $P(e, 2)$. Show that the acute

angle between these curves is given by $\tan \theta = \frac{4e}{e^2 - 4}$. [4]

c)

i) Show that $\frac{x+2}{x+3} = 1 - \frac{1}{x+3}$ [2]

ii) Hence find the value of the definite integral $\int_{-2}^0 \frac{x+2}{x+3} dx$ [2]

d) Prove that $e^{\ln a} = a$ and hence determine the value of $\int_{-\ln 2}^{\ln 2} e^x dx$ [4]

e) Find the second derivative of e^{5x^2} . Hence show that the curve $y = e^{5x^2}$ is concave up for all real values of x . [3]

Question 2 (17 marks)

a) Find $\int \frac{t}{\sqrt{1+t}} dt$ using $u = 1+t$ [3]

b) Find $\int \cos^2 4x \, dx$ [3]

c) Find $\int_0^1 x(5x^2 - 4) \, dx$ using the substitution $u = 5x^2 - 4$ [3]

d) Evaluate $\int_0^{\frac{\pi}{3}} \tan x \, dx$ using the substitution $u = \cos x$. [4]

e) Use the substitution $u = x^4 + 4x + 1$ to evaluate to 2 decimal places.

$$\int_0^1 (x^3 + 1) \sqrt[4]{x^4 + 4x + 1} \, dx$$
 [4]

Question 3 (19 marks)

a)

i) Sketch the graph of $x = \sin y$ and on it clearly mark the portion taken

as $y = \sin^{-1} x$. [1]

ii) State the domain and range of the function $y = \sin^{-1} 2x$. [2]

b) What is the value of $\tan^{-1}(\sqrt{3}) - \tan^{-1}(-1)$. [2]

c) Find the derivative of

i) $\sin^{-1}(\cos x)$ [2]

ii) $\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{x}\right)$ [3]

iii) $x \log_e(\cos^{-1} x)$ [3]

d) Evaluate these integrals

i) $\int \frac{1}{5\sqrt{16-x^2}} dx$ [1]

ii) $\int \frac{dx}{x^2 + 2x + 10}$ [2]

iii) $\int_{-3}^0 \frac{dt}{t^2 + 9}$ [3]

QUESTION 1 (17 marks)

$$a) \frac{e^x}{e^x + e^{-x}} = \frac{e^x}{e^x + e^{-x}} \times \frac{e^x}{e^x}$$

$$= \frac{e^{2x}}{e^{2x} + 1}$$

b) $y = \frac{2e}{x}$ and $y = \log_e x^2$ intersect
at $P(e, 2)$

$$y' = -2ex^{-2}$$

at $x = e, y' = \frac{-2e}{e^2} = \frac{-2}{e}$

$$y' = \frac{2}{x}$$

at $x = e, y' = \frac{2}{e}$

✓ for both gradients

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-\frac{2}{e} - \frac{2}{e}}{1 + \left(-\frac{2}{e}\right)\left(\frac{2}{e}\right)} \right|$$

$$= \left| \frac{-\frac{4}{e}}{1 - \frac{4}{e^2}} \right|$$

$$= \left| \frac{-\frac{4}{e}}{\frac{e^2 - 4}{e^2}} \right|$$

$$= \left| -\frac{4}{e} \times \frac{e^2}{e^2 - 4} \right|$$

$$= \left| \frac{-4e}{e^2 - 4} \right|$$

$$= \frac{4e}{e^2 - 4}$$

Mostly well
calculated

Some students
did not sub.
 $x = e$ at this
stage which
resulted in
baseless manipulations
Some used the
quotient rule
for $y = \frac{2e}{x}$
ineffectively

-1 mark for
not using
absolute value

$$\frac{x+3}{x+3} = \frac{x+3}{x+3}$$

$$= \frac{x+3}{x+3} - \frac{1}{x+3}$$

$$= 1 - \frac{1}{x+3}$$

ii) $\int_{-2}^0 \frac{x+2}{x+3} dx = \int_{-2}^0 \left(1 - \frac{1}{x+3}\right) dx$

$$= \left[x - \ln(x+3) \right]_{-2}^0$$

$$= 0 - \ln 3 - (-2) - \ln 1$$

$$= 2 - \ln 3$$

d) $y = e^{\ln a}$

$$\ln y = \ln e^{\ln a}$$

$$\ln y = \ln a \ln e$$

$$\ln y = \ln a$$

$$y = a$$

$$\therefore e^{\ln a} = a$$

$$\int_{-\ln 2}^{\ln 2} e^x dx = \left[e^x \right]_{-\ln 2}^{\ln 2}$$

$$= e^{\ln 2} - e^{-\ln 2}$$

$$= 2 - e^{\ln 2^{-1}}$$

$$= 2 - \frac{1}{2}$$

$$= \frac{3}{2}$$

mostly well
done

some did not
use $\ln 1 = 0$

Very poorly done
by many
students either
not showing
enough detail
or working
on both
sides at same
time

Some did
not realize
 $e^{\ln 2^{-1}} = e^{\ln \frac{1}{2}}$
 $= \frac{1}{2}$

e) $y = e^{5x^2}$
 $y' = 10x e^{5x^2}$
 $y'' = 10x \times 10x e^{5x^2} + e^{5x^2} \times 10$
 $= 10e^{5x^2}(10x^2 + 1)$

$y'' > 0$ for all values of x so
the curve is always concave up. ✓

Question 2 (17 marks)

a) $\int \frac{t}{\sqrt{1+t}} dt$: $u = 1+t$, $t = u-1$
 $du = dt$ ✓

$$\int \frac{u-1}{\sqrt{u}} du$$

$$\int (u^{\frac{1}{2}} - u^{-\frac{1}{2}}) du$$

$$\frac{2u^{\frac{3}{2}}}{\frac{3}{2}} - 2u^{\frac{1}{2}} + C$$

$$\frac{2(1+t)^{\frac{3}{2}}}{3} - 2\sqrt{1+t} + C$$

b) $\int \cos^2 4x dx = \frac{1}{2} \int \cos 8x + 1 dx$ ✓

$$= \frac{1}{2} \left(\sin \frac{8x}{8} + x \right) + C$$

$$= \frac{\sin 8x}{16} + \frac{x}{2} + C$$

(3)

Many did not find y'' accurately
some did not conclude appropriately

Mostly well done
Note

$$\frac{1}{\sqrt{u}} = u^{-\frac{1}{2}}$$

Some people left in u form

Well done

Some mistakes with substitution

$$\cos^2 4x = \frac{1}{2}(1 + \sin 8x)$$

c) $\int_0^1 x(5x^2 - 4) dx$

$$u = 5x^2 - 4 \quad \text{if } x=0, u=-4$$

$$du = 10x dx \quad x=1, u=1$$

$$\frac{1}{10} \int_{-4}^1 u du$$

$$\frac{u^2}{20} \Big|_{-4}^1$$

$$\frac{1}{20} - \frac{16}{20}$$

$$-\frac{3}{4}$$

d) $\int_0^{\frac{\pi}{3}} \tan x dx$

$$u = \cos x$$

$$du = -\sin x dx$$

$$\text{if } x=0, u=1$$

$$x=\frac{\pi}{3}, u=\frac{1}{2}$$

$$\int_0^{\frac{\pi}{3}} \frac{\sin x}{\cos x} dx$$

$$\int_1^{\frac{1}{2}} -\frac{1}{u} du$$

$$-\ln u \Big|_1^{\frac{1}{2}}$$

$$-\ln \frac{1}{2} - (-\ln 1)$$

$$\ln 1 - \ln \frac{1}{2}$$

$$0 + \ln 2$$

$$\ln 2$$

e) $\int_0^1 (x^3 + 1) \sqrt{x^4 + 4x + 1} dx$

$$u = x^4 + 4x + 1$$

$$du = 4x^3 + 4$$

$$= 4(x^3 + 1)$$

$$\text{if } x=0, u=1$$

$$x=1, u=6$$

$$\frac{1}{4} \int_1^6 u^{\frac{1}{2}} du$$

$$\frac{4u^{\frac{3}{4}}}{\frac{4}{3}} \Big|_1^6$$

$$\frac{1}{5} \left(6^{\frac{5}{4}} - 1^{\frac{5}{4}} \right)$$

$$1.678101496$$

$$1.68 \text{ (to 2 d.p.)}$$

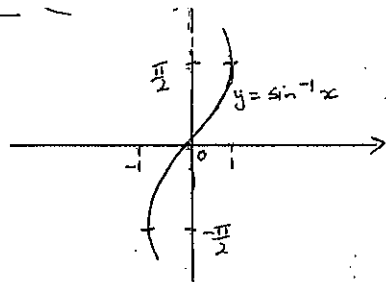
Well done

Some mistakes with

$$-\int \frac{1}{u} du = -\ln u$$

Well done Overall

a) i)



ii) Domain $-1 \leq x \leq 1$

$$\{x: -\frac{1}{2} \leq x \leq \frac{1}{2}\}$$

$$\text{Range } \{-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}\}$$

b) $\tan^{-1}(\sqrt{3}) - \tan^{-1}(-1)$

$$x = \tan^{-1} \sqrt{3}$$

$$\tan x = \sqrt{3}$$

$$x = \frac{\pi}{3}$$

$$y = \tan^{-1}(-1)$$

$$\tan y = -1$$

$$y = -\frac{\pi}{4}$$

$$\begin{aligned} \tan^{-1}(\sqrt{3}) - \tan^{-1}(-1) &= \frac{\pi}{3} - -\frac{\pi}{4} \\ &= \frac{7\pi}{12} \end{aligned}$$

c) i) $\frac{d}{dx} \sin^{-1}(\cos x) = \frac{1}{\sqrt{1-\cos^2 x}} \times -\sin x$

$$= \frac{-\sin x}{\sin x}$$

$$= -1$$

c) Some students graphed $y = \sin x$ by mistake

a) well done

b) Some

students used

$$\tan(x+y)$$

$$= \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

which did not lead to a result

c) i) well done

$$\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{1}{1+x^2} + \frac{1}{1+x^2} \times -x^{-2}$$

$$\frac{1}{1+x^2} - \frac{1}{x^2} = \frac{x^2+1}{x^2}$$

$$\frac{1}{1+x^2} - \frac{1}{1+x^2}$$

$$0$$

iii) $\frac{d}{dx} (x \log_e(\cos^{-1} x))$

$$x \times \frac{-1}{\sqrt{1-x^2}} \times \cos^{-1} x + \log_e(\cos^{-1} x) \times 1$$

$$\frac{-x}{(1-x^2)\cos^{-1} x} + \log_e(\cos^{-1} x)$$

d) i) $\int \frac{1}{5\sqrt{16-x^2}} dx = \frac{1}{5} \sin^{-1} \frac{x}{4} + c$

ii) $\int \frac{dx}{x^2+2x+10} = \int \frac{dx}{(x+1)^2+9}$

$$= \frac{1}{3} \tan^{-1} \left(\frac{x+1}{3} \right) + c$$

iii) $\int_{-3}^0 \frac{dt}{t^2+9} = \frac{1}{3} \left[\tan^{-1} \frac{t}{3} \right]_{-3}^0$

$$= \frac{1}{3} \left[\tan^{-1} 0 - \tan^{-1} -1 \right]$$

$$= \frac{1}{3} \left(0 - -\frac{\pi}{4} \right)$$

$$= \frac{\pi}{12}$$

(6)

Some students failed to use chain rule to find $\frac{d}{dx} \left(\frac{1}{x} \right)$

ii) well done

d) i) well done

a) Many students

failed to factor $x^2+2x+10 = (x+1)^2+9$,

and hence tried to use logs unsuccessfully.

iii) well done